

Pointers Practice Problems

1. Explain the difference between a pointer and a regular variable in C.
2. What is the purpose of the & and * operators in pointer manipulation?
3. How is pointer arithmetic performed in C? Explain with examples.
4. Predict the output for each blank cell based on the code and initial assumptions.

Assume:

- Address of `x` = **1000**
- Address of `y` = **2000**
- Address of `z` = **3000**

Initial Values:

```
int x = 10, y = 20, z = 5;  
int *p = &x, *q = &y, *r = &z;
```

Step	Code	x	y	z	p	q	r
	-	10	20	5	1000	2000	3000
1.	<code>*p = *q + z;</code>						
2.	<code>q = p;</code>						
3.	<code>*q = *q - *r;</code>						
4.	<code>r = q;</code>						
5.	<code>*r = *p + *q;</code>						
6.	<code>p = &z;</code>						
7.	<code>*p = *p * 2;</code>						
8.	<code>p = &y;</code>						
9.	<code>*p = *q / *r;</code>						
10.	<code>r -= 2;</code>						

5. Write a program to swap two integers using pointers.
6. Given an array of exam scores, use pointers to find the top 3 scores and their positions in the array.
7. Write a program that checks if two strings are identical using pointers.

8. Write a program that checks if a given string is URL which starts with "http" or "https" using pointers.
9. Write a program to sort an array of integers in ascending order using pointer manipulation.
10. Write a program that rotates the elements of an array to the right by k positions using pointers.
11. Write a program to search for a key element in an unsorted array using pointers.
12. Write a program to concatenate two strings using pointers.
13. Implement a C program to reverse an integer array in place using pointers.
14. Write a program to count the number of vowels in a string using pointers.